

**Ahmedabad
University**

Milestone 1(M1): Project Proposal and Scoping

Breast Cancer Survival Prediction

CSE516: Probabilistic Graphical Models

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1 Domain and Problem Definition

1.1 Domain

The proposed project focuses on the healthcare domain, specifically breast cancer survival prediction using multimodal patient data. The project considers two complementary data modalities: genomic data and histology-derived image data.

Genomic data provides molecular information about the cancer, while histology-derived data provides information obtained from tissue images. The project investigates how these two modalities can be jointly represented using a probabilistic graphical model (PGM).

1.2 Problem Definition

The problem addressed in this project is the joint modelling of genomic and histology-derived data for breast cancer survival prediction.

Let X denote the genomic observations and Y denote the histology-derived observations. The project considers a latent variable Z that represents the shared underlying information between the two modalities.

The probabilistic graphical model represents the dependencies between these variables as

$$Z \rightarrow X, \quad Z \rightarrow Y. \quad (1)$$

This representation assumes that the two observed modalities are conditionally independent given the latent variable Z :

$$X \perp Y \mid Z. \quad (2)$$

The project investigates this latent-variable representation for multimodal data and studies how the model can be learned and used for inference in the context of breast cancer survival prediction.

1.3 Motivation and Significance

Breast cancer survival prediction can involve multiple types of patient data. Genomic and histology-derived data provide complementary information about the same cancer, and jointly modelling these modalities can help capture information that may not be represented by either modality alone.

The reference work investigates the use of canonical correlation analysis (CCA) to obtain joint embeddings of histology and genomic data for breast cancer survival prediction. In particular, it studies whether a shared latent representation can capture relationships between the two modalities and provide useful information for downstream survival prediction.

The motivation for using a probabilistic graphical model in this project is to explicitly represent the relationship between the observed modalities and the latent variable. This allows the project to study not only the final survival prediction, but also the three central PGM components: representation of the dependencies, learning of the model parameters, and inference of the latent variable.

Therefore, the project will investigate whether an alternative parameter-learning approach can be used within the same latent-variable PGM and how the choice of learning method affects the learned latent representation and its subsequent use for survival prediction.

1.4 Assumptions

The proposed probabilistic model is based on the assumption that the genomic and histology-derived observations provide different views of the same underlying breast cancer. These observations are therefore modelled through a shared latent variable Z .

The main assumptions considered in the project are:

1. The genomic observations X and histology-derived observations Y are generated from a common latent variable Z .
2. The two modalities are conditionally independent given the latent variable. Thus,

$$X \perp Y \mid Z. \tag{3}$$

3. The latent variable provides a lower-dimensional representation of information shared across the two modalities.
4. The same underlying probabilistic graphical model is used when comparing the original CCA-based parameter learning with the proposed EM-based parameter learning.

These assumptions define the probabilistic structure that will be investigated rather than introducing a new graphical structure during the project.

1.5 Scope of the Project

The project is restricted to one healthcare application: multimodal breast cancer survival prediction using genomic and histology-derived data.

The project will use the processed TCGA-BRCA data associated with the reference work. The dataset contains genomic and histology/image-derived features for the same patients, allowing the two modalities to be modelled jointly through the latent variable Z .

The first part of the project will reproduce the CCA-based parameter learning approach from the reference work. This reproduced model will serve as the baseline for the subsequent experiments.

The main extension of the project is to investigate Expectation-Maximization (EM) as an alternative parameter-learning method for the same latent-variable PGM. The learned models will then be used for probabilistic inference of the latent variable from multimodal and modality-specific observations.

The project will investigate the following settings:

- genomic and histology-derived observations available together;
- genomic observations available without histology-derived observations;
- histology-derived observations available without genomic observations; and

- comparison of CCA-based and EM-based parameter learning under these settings.

The evaluation will focus on the learned latent representation, probabilistic inference, and downstream breast cancer survival prediction. The primary survival-prediction measure planned for the project is the concordance index (C-index), together with measures of EM convergence and changes in latent-state inference under incomplete modalities.

The project does not aim to introduce a completely new graphical model structure. Instead, it investigates whether an alternative parameter learning strategy, combined with explicit latent-state inference, provides useful technical insight within the existing multimodal PGM.

2 Mapping the Problem onto Representation, Inference and Learning

The proposed project investigates the use of a probabilistic graphical model (PGM) to jointly represent genomic and histology information for breast cancer survival prediction. The model is organized around the three core PGM components required for the project: **Representation**, **Inference**, and **Learning**. Representation defines the random variables and their probabilistic dependencies, inference specifies what quantities will be computed from the model, and learning determines how the model parameters will be estimated from the available patient data.

2.1 Representation

2.1.1 Random Variables and Nodes

The proposed model represents the two observed data modalities through a shared latent variable. For each patient, let

- X denote the observed genomic feature vector,
- Y denote the observed histology/image feature vector, and
- Z denote the latent shared representation connecting the two modalities.

Thus, the set of variables represented in the graphical model is

$$\mathcal{V} = \{Z, X, Y\}.$$

The variables X and Y are observed, whereas Z is unobserved and must be estimated from the observed multimodal data. The latent variable Z is intended to capture the shared structure between genomic and histology information that can subsequently be used for survival prediction.

It is important to distinguish the latent variable Z from the survival outcome. The variable Z does not represent the observed survival time or event indicator. Instead, it represents a lower-dimensional latent structure learned from the two observed modalities. The resulting latent representation can then be used as an input to a downstream survival prediction model.

2.1.2 Graph Structure

The initial graphical structure of the proposed model is

$$Z \longrightarrow X, \quad Z \longrightarrow Y.$$

The corresponding PGM is shown in Figure 2.3.2.

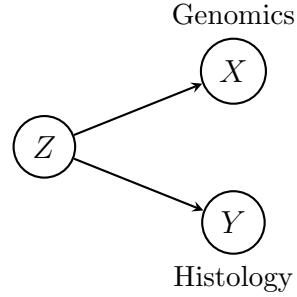


Figure 1: Probabilistic graphical model representing the shared latent variable Z and its dependencies on the observed genomic (X) and histology (Y) modalities.

As shown in Figure 2.3.2, the latent variable Z is the common parent of the two observed modalities. The two directed edges represent the dependency of the genomic and histology observations on the shared latent representation.

2.1.3 Probabilistic Dependencies

The graph implies the following factorization of the joint probability distribution:

$$p(Z, X, Y) = p(Z) p(X | Z) p(Y | Z).$$

The corresponding conditional independence assumption is

$$X \perp Y | Z.$$

This means that genomic observations and histology observations are modeled as conditionally independent once the shared latent variable Z is known. The dependence between the two modalities is therefore captured through their common latent representation.

This dependency structure is appropriate for the problem because genomics and histology represent different types of measurements obtained from the same patient. They exist in different feature spaces, but the project is interested in the information that is shared between them. A latent-variable PGM provides an explicit probabilistic mechanism for representing this shared information.

Rather than directly assuming that all genomic and histology features should be treated as a single feature vector, the model introduces Z as an intermediate representation through which the two modalities are related. This provides a probabilistic interpretation of multimodal integration and allows the latent representation to be inferred even when only a subset of the modalities is available.

2.1.4 Linear-Gaussian Model

The proposed representation follows the linear-Gaussian latent-variable formulation underlying probabilistic canonical correlation analysis (pCCA). The latent variable is assigned a standard multivariate Gaussian prior:

$$Z \sim \mathcal{N}(0, I_d),$$

where d denotes the dimensionality of the latent representation.

Conditioned on Z , the genomic observations are modeled as

$$X | Z \sim \mathcal{N}(W_X Z, \Psi_X),$$

while the histology observations are modeled as

$$Y | Z \sim \mathcal{N}(W_Y Z, \Psi_Y).$$

Here, W_X and W_Y are loading matrices that map the latent representation to the genomic and histology feature spaces, respectively. The matrices Ψ_X and Ψ_Y represent modality-specific residual covariance.

The complete generative model is therefore

$$p(Z, X, Y) = p(Z) p(X | Z) p(Y | Z).$$

This formulation allows the observed genomic and histology measurements to be represented as different manifestations of a common latent structure, while still allowing each modality to

contain variation that is specific to that modality.

2.1.5 Why the Representation is Appropriate

The proposed representation is appropriate for the project for three main reasons.

First, the two modalities contain complementary information about the same patient but are represented in substantially different feature spaces. A shared latent variable provides a common space in which information from both modalities can be represented.

Second, the pCCA-style linear-Gaussian formulation provides an explicit probabilistic model rather than treating multimodal integration solely as feature concatenation. The model explicitly represents the conditional relationship between the two modalities through Z .

Third, because Z is latent, the model naturally supports inference of the shared representation from the available observations. This makes it possible to investigate both the normal setting in which both modalities are available and settings in which one modality is missing.

2.2 Inference

2.2.1 Inference Objective

Since Z is an unobserved variable, it cannot be directly obtained from the dataset. The inference stage therefore estimates the posterior distribution of the latent variable given the observed modalities.

When both genomic and histology observations are available, the primary inference quantity is

$$p(Z \mid X, Y).$$

Using Bayes' rule,

$$p(Z \mid X, Y) \propto p(X, Y \mid Z)p(Z).$$

Under the conditional-independence assumption of the graphical model,

$$p(Z \mid X, Y) \propto p(X \mid Z)p(Y \mid Z)p(Z).$$

The resulting posterior distribution describes the latent shared representation that is most consistent with the observed genomic and histology data.

For downstream use, a point estimate of the latent representation can be obtained using the posterior mean:

$$\hat{Z}_{XY} = \mathbb{E}[Z \mid X, Y].$$

This inferred representation will be used as the multimodal representation for the downstream survival prediction task.

2.2.2 Inference with Both Modalities

The primary inference setting is the case where both modalities are available. The inference process can be represented as

$$(X, Y) \longrightarrow p(Z \mid X, Y) \longrightarrow \hat{Z}_{XY}.$$

The inferred representation $\hat{Z}_{XY}$ contains the information from both genomic and histology observations that is represented through the learned latent structure.

The resulting representation will then be passed to the survival prediction stage, where its usefulness for predicting patient survival will be evaluated.

2.2.3 Inference with a Missing Modality

The latent-variable formulation also permits investigation of cases where one of the two modalities is unavailable.

If the histology modality is unavailable, inference can be performed from the genomic observations alone:

$$p(Z \mid X).$$

The corresponding inference process is

$$X \longrightarrow p(Z \mid X) \longrightarrow \hat{Z}_X.$$

Similarly, if genomic information is unavailable, the model can infer the latent representation from histology observations:

$$p(Z \mid Y),$$

giving

$$Y \longrightarrow p(Z \mid Y) \longrightarrow \hat{Z}_Y.$$

These cases will allow the project to investigate whether the learned probabilistic representation can still provide useful latent information when one of the modalities is missing.

2.2.4 Inference Method

Because the proposed model is linear-Gaussian, the posterior distribution of the latent variable can be obtained analytically once the model parameters have been learned. Therefore, the basic complete-modality inference problem does not require a separate approximate inference method.

The learned parameters

$$\Theta = \{W_X, W_Y, \Psi_X, \Psi_Y\}$$

define the posterior distribution of Z for a given patient's observed data. The posterior mean or an equivalent MAP estimate can then be used as the patient's inferred latent representation.

Thus, the inference stage will compute

$$\hat{Z} = \mathbb{E}[Z \mid \text{available observations}],$$

where the available observations may be (X, Y) , X alone, or Y alone.

2.3 Learning

2.3.1 Learning Objective

The graphical structure defines the relationships between the variables, but the numerical parameters of the model must be learned from the available patient data.

The main model parameters are

$$\Theta = \{W_X, W_Y, \Psi_X, \Psi_Y\}.$$

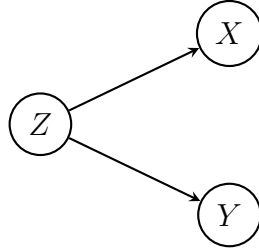
The learning objective is to estimate these parameters so that the linear-Gaussian latent-variable model provides an appropriate probabilistic description of the observed genomic and histology data.

The project therefore focuses primarily on **parameter learning**.

2.3.2 Structure Learning

Structure learning is not the primary learning task in this project.

The initial graph structure:



is specified according to the latent-variable formulation underlying pCCA. We therefore begin with a predefined graphical structure rather than searching over alternative graph structures.

The rationale is that the project starts with a specific hypothesis: the genomic and histology modalities are related through a common latent representation. The purpose of the project is to estimate the parameters of this model and investigate the resulting latent representation, rather than to discover an arbitrary graphical structure from the data.

Accordingly,

Structure learning: not the primary learning task

and

Parameter learning: primary learning task

.

Structure learning may be considered as a possible future extension, but it is outside the core scope of the proposed M1 project.

2.3.3 CCA-Based Baseline

The existing CCA-based approach will be used as the primary baseline for comparison with the proposed probabilistic parameter-learning approach.

Canonical correlation analysis (CCA) learns projection matrices U and V for the two modalities such that the projected representations have high cross-modal correlation. The two projected representations are

$$XU \quad \text{and} \quad YV.$$

The resulting representations can be combined as

$$H_{\text{CCA}} = [XU; YV].$$

The CCA baseline provides a reference for evaluating the proposed probabilistic learning approach. It also establishes a connection with the existing implementation and the source work on multimodal breast cancer survival prediction.

The baseline is therefore not a separate graphical structure. Instead, it provides a comparison against which the parameter-learning approach for the latent-variable PGM can be assessed.

2.3.4 Proposed EM-Based Parameter Learning

The proposed extension is to learn the parameters of the same linear-Gaussian latent-variable model using **Expectation-Maximization (EM)**.

EM is appropriate because the model contains an unobserved latent variable Z . The observed data consist of X and Y , while the latent values of Z must be estimated during learning.

The EM algorithm alternates between an Expectation step and a Maximization step.

Expectation step. Given the current parameter estimates $\Theta^{(t)}$, the E-step computes the posterior distribution of the latent variable:

$$p(Z \mid X, Y; \Theta^{(t)}).$$

The posterior expectations required for parameter estimation are then computed. In particular, the procedure uses quantities such as

$$\mathbb{E}[Z \mid X, Y]$$

and

$$\mathbb{E}[ZZ^T \mid X, Y].$$

These expectations provide estimates of the unobserved latent quantities under the current model parameters.

Maximization step. In the M-step, the model parameters are updated using the expectations obtained from the E-step. The update seeks to maximize the expected complete-data log-likelihood:

$$\Theta^{(t+1)} = \arg \max_{\Theta} \mathbb{E}_{Z \mid X, Y; \Theta^{(t)}} [\log p(X, Y, Z; \Theta)].$$

The parameter updates produce new estimates of

$$W_X, \quad W_Y, \quad \Psi_X, \quad \Psi_Y.$$

The E-step and M-step are then repeated until a predefined convergence criterion is satisfied. The overall EM learning process is therefore

Observed $(X, Y) \longrightarrow$ Parameter initialization $\longrightarrow$ *E*-step $\longrightarrow$ *M*-step $\longrightarrow$ Convergence.

The final learned parameters are then used during inference to estimate the latent representation for individual patients.

2.3.5 Parameter Learning versus Structure Learning

The distinction between the two types of learning is important for the scope of this project.

- **Structure learning** determines the graphical structure, such as which variables are connected by edges. In this project, the initial structure is specified as $Z \rightarrow X$ and $Z \rightarrow Y$.
- **Parameter learning** estimates the numerical parameters associated with the specified structure. In this project, the primary parameter-learning approach will be EM, with CCA serving as the comparison baseline.

Thus, the project does not require a structure-search procedure as part of its core learning pipeline. The main learning problem is estimating the parameters of the predefined latent-variable PGM.

2.4 Integrated Representation, Inference and Learning Pipeline

The three PGM verticals form a single connected pipeline rather than three independent components.

The proposed workflow begins with the predefined graphical representation:

$$Z \rightarrow X, \quad Z \rightarrow Y.$$

The model parameters are then obtained from data using the learning stage. The existing CCA formulation provides the baseline, while EM provides the proposed parameter-learning approach.

Once the parameters have been learned, inference is performed to obtain the latent representation from the available observations. For complete multimodal observations this corresponds to

$$p(Z \mid X, Y),$$

while missing-modality settings use

$$p(Z \mid X)$$

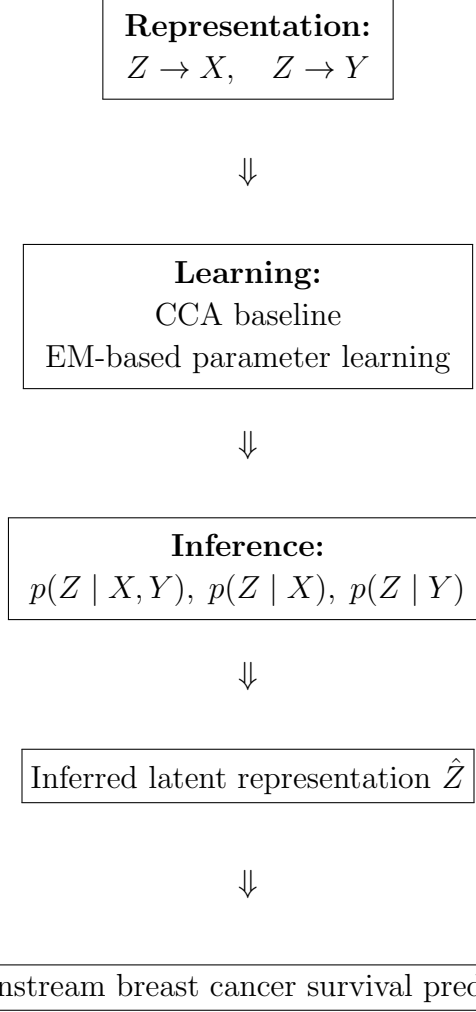
or

$$p(Z | Y).$$

The inferred latent representation is then used for downstream survival prediction. The complete conceptual mapping is therefore

$$\boxed{\text{Representation} \rightarrow \text{Learning} \rightarrow \text{Inference} \rightarrow \text{Survival Prediction}}$$

with the individual components defined as



This Representation–Learning–Inference mapping establishes the conceptual backbone of the proposed project. The representation defines how genomic and histology observations are probabilistically related through a shared latent variable. Parameter learning obtains the numerical parameters of this model from patient data, while inference computes the latent representation for new or partially observed patients. The resulting representation is then evaluated through the downstream survival prediction task.

3 Establishing the State-of-the-Art (SOTA) Position

3.1 Selected SOTA Reference Work

For the initial SOTA positioning, we consider the recent work by Subramanian, Syeda-Mahmood, and Do, titled “*Modelling-based joint embedding of histology and genomics using canonical correlation analysis for breast cancer survival prediction.*” The work was published in *Artificial Intelligence in Medicine*, Volume 149, in 2024. One of the authors, Tanveer Syeda-Mahmood, is affiliated with IBM Research Almaden, while the other authors are affiliated with the University of Illinois at Urbana-Champaign. The paper is particularly relevant to our project because it addresses the same application problem, uses the same multimodal setting of genomic and histology data, and formulates their relationship using a probabilistic graphical model (PGM).

The paper investigates whether multimodal breast cancer data can be explicitly modeled as originating from a common underlying cancer source. Rather than treating genomic and histology observations as independent inputs, the authors introduce a shared latent variable through which the two modalities are probabilistically related. The authors then obtain maximum-likelihood estimates of the PGM parameters using canonical correlation analysis (CCA) and use the resulting representation for downstream survival prediction.

This work is therefore selected as the principal reference for our SOTA positioning because it provides the direct methodological foundation for the proposed project. Our project does not introduce a completely different graphical model; instead, it reproduces this formulation and investigates an alternative parameter-learning procedure together with explicit latent-state inference and incomplete-modality experiments.

3.2 Problem Addressed

The selected work addresses the problem of breast cancer survival prediction using multiple data modalities, specifically genomic and histology information. Breast cancer exhibits substantial heterogeneity, and different modalities capture different characteristics of the same patient’s cancer. The authors therefore investigate whether these modalities can be jointly represented by explicitly modeling them as originating from a common latent source.

The central problem can be expressed as the integration of two observed modalities,

$$X = \text{genomic information}, \quad Y = \text{histology information},$$

through a shared latent variable Z .

The motivation for this formulation is that the two modalities provide different but related views of the same patient. The paper aims to exploit their shared information for downstream prediction while retaining a probabilistic interpretation of the multimodal relationship.

This problem formulation directly corresponds to our project objective. Our proposal adopts the same multimodal breast cancer survival problem and uses the same processed TCGA-BRCA data associated with the base work. Our main extension is to investigate EM-based parameter learning for the same latent-variable PGM and to explicitly study latent-state inference and incomplete modalities.

3.3 Dataset Used in the Reference Work

The selected work evaluates its method on the **TCGA-BRCA (The Cancer Genome Atlas Breast Invasive Carcinoma)** dataset. The processed dataset used in the study contains 974 patients with both genomic and histology-derived information.

For the histology modality, whole-slide images were processed into patches and CellProfiler was used to obtain image-derived features. The extracted features were summarized across patches using five-bin histograms, resulting in 1075-dimensional imaging feature vectors for each patient.

For the genomic modality, RNA-sequencing expression data containing more than 20,000 genes were considered. The expression values were log-transformed and the most variable genes were selected using the coefficient of variation. The top 1000 genes were retained, resulting in a 1000-dimensional genomic feature vector for each patient.

Therefore, the two main observed data matrices can be represented as

$$X \in \mathbb{R}^{N \times 1000}, \quad Y \in \mathbb{R}^{N \times 1075},$$

where N denotes the number of patients.

The study randomly divides the data into 60%-15%-25% training, validation, and testing sets and generates five data splits. Survival information is handled in the downstream survival prediction stage using the Cox proportional hazards model and Kaplan–Meier estimation.

These same processed TCGA-BRCA genomic and histological data are specified in our project proposal as the data source for reproducing the original PGM, implementing EM-based learning, performing latent-state inference, and evaluating incomplete-modality settings.

3.4 Representation and Probabilistic Model

The central representation used in the selected work is a two-modality latent-variable probabilistic graphical model. The latent variable Z influences both observed modalities X and Y :

$$Z \longrightarrow X, \quad Z \longrightarrow Y.$$

The corresponding graphical structure is shown in Figure 2.

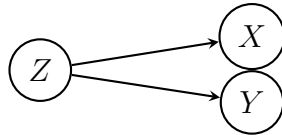


Figure 2: Latent-variable PGM used in the selected reference work, where the shared latent variable Z connects the genomic and histology modalities.

The model assigns a Gaussian prior to the latent variable:

$$Z \sim \mathcal{N}(0, I_d),$$

and models the two observed modalities conditionally on Z as

$$X \mid Z \sim \mathcal{N}(W_X Z, \Psi_X),$$

$$Y \mid Z \sim \mathcal{N}(W_Y Z, \Psi_Y).$$

Consequently, the joint distribution factorizes as

$$p(Z, X, Y) = p(Z)p(X \mid Z)p(Y \mid Z).$$

The graphical model therefore implies the conditional independence

$$X \perp Y \mid Z.$$

In the breast cancer setting, the latent variable represents the underlying shared information associated with the patient’s cancer and survival potential, while X and Y correspond to genomic and histology observations, respectively.

This representation is also the graphical structure adopted in our project. Thus, the proposed work is an extension of the parameter-learning and inference procedures rather than a change to the fundamental PGM structure.

3.5 Learning Methodology

The selected work obtains estimates of the parameters of the latent-variable model using canonical correlation analysis (CCA). CCA identifies projection directions for the two modalities that maximize their cross-modal correlation. If U and V denote the corresponding projection matrices, the resulting modality-specific representations are

$$XU \quad \text{and} \quad YV.$$

These representations can subsequently be combined into a joint multimodal representation:

$$H_{\text{CCA}} = [XU ; YV].$$

The probabilistic interpretation of CCA allows the CCA solutions to be related to maximum-likelihood parameter estimates for the probabilistic latent-variable formulation.

Because the genomic and histology data are high-dimensional, the study also investigates penalized CCA variants. In particular, sparse CCA (SCCA) and graph-informed variants are considered to incorporate properties such as sparsity and feature relationships. For generating multiple embedding dimensions, the work investigates three deflation strategies:

- Hotelling Deflation (HD),
- Projected Deflation (PD), and
- Orthogonalized Projected Deflation (OPD).

These procedures are used to obtain richer multimodal embeddings while encouraging additional embedding directions to capture different cross-modal information.

3.6 Inference Methodology

After estimating the parameters of the probabilistic model, the selected work considers inference of the latent variable from the observed modalities. When both modalities are available, the latent representation can be obtained from the posterior distribution

$$p(Z \mid X, Y).$$

The posterior mean provides an estimate of the latent variable:

$$\hat{Z} = \mathbb{E}[Z \mid X, Y].$$

Thus, the reference work connects the observed multimodal data to the latent representation before using the representation for downstream prediction.

The selected work also presents the equivalence between obtaining CCA-based joint embeddings and using the probabilistic latent-variable formulation for prediction. In the survival prediction setting, the learned multimodal representation is supplied to a downstream survival predictor.

This inference formulation is particularly relevant to our project because our proposal makes latent-state inference an explicit project objective. In addition to multimodal inference,

$$P(Z \mid X, Y),$$

we plan to investigate modality-specific inference:

$$P(Z \mid X)$$

and

$$P(Z \mid Y).$$

Where applicable, MAP inference will also be considered:

$$Z_{\text{MAP}} = \arg \max_Z P(Z \mid X, Y).$$

Therefore, the proposed project makes the latent inference component more explicit and extends the investigation to incomplete-modality conditions.

3.7 Evaluation Methodology

The selected work evaluates the proposed multimodal representation through both simulated experiments and real-world breast cancer survival prediction.

3.7.1 Simulation Evaluation

In the simulation experiments, the authors construct data for which the underlying latent variable is known. This allows them to directly evaluate how accurately the learned multimodal representation recovers the hidden variable.

The mean squared error (MSE) between the ground-truth latent variable and the predicted latent variable is used to evaluate latent-variable prediction.

The simulations also evaluate the ability of different CCA variants and deflation schemes to identify additional cross-modal correlations.

3.7.2 TCGA-BRCA Survival Evaluation

For the real breast cancer dataset, the learned multimodal embeddings are used for survival prediction through a Cox proportional hazards model with elastic-net regularization.

The primary survival metric is the concordance index (C-index), which measures the agreement between predicted patient risk ordering and observed survival outcomes.

The study also uses Kaplan–Meier survival analysis to investigate whether patients separated using the learned embeddings exhibit different survival patterns.

The evaluation therefore connects the learned multimodal representation to a clinically relevant downstream task rather than evaluating the latent representation only in isolation.

3.8 Findings of the Reference Work

The reference work reports that the pCCA-based joint embeddings can provide useful multimodal representations for breast cancer survival prediction.

For the TCGA-BRCA experiments, the PCA-based concatenated representation achieved an average C-index of

$$66.47 \pm 7.66\%.$$

Among the SCCA configurations, the concatenated representation obtained using the Orthogonalized Projected Deflation (OPD) scheme achieved the reported best C-index of

$$68.32 \pm 6.69\%.$$

The study also reports that the genomic modality generally provides stronger survival-prediction performance than the histology modality when considered individually.

The results indicate that multimodal representations obtained through the pCCA framework can capture useful cross-modal information and can be used for downstream survival prediction. The work also demonstrates that different deflation strategies influence the quality of the resulting multidimensional embeddings.

These values are reported results from the reference work and are not results obtained by our project. The corresponding reproduction and comparison of the CCA-based baseline will form part of our later experimental work.

3.9 Limitations and Open Directions

The reference work identifies several limitations and possible directions for further investigation.

First, the histology features used in the study are based on relatively simple CellProfiler-derived features. The authors note that richer image features or deep-learning-based feature extraction could potentially provide more informative representations.

Second, the graph-informed and sparse variants of CCA can become computationally challenging in the high-dimensional setting. The study reports computational instability associated with high-dimensional matrix operations for some configurations.

Third, the survival analysis based on clustering of the learned embeddings does not establish statistically significant separation for all considered embeddings. The authors therefore indicate that the clustering strategy and the number of clusters could be investigated further.

Fourth, the model is based on linear CCA relationships. The authors identify kernel CCA and deep CCA as possible directions for capturing nonlinear relationships between modalities.

Finally, the paper indicates that comparisons with other state-of-the-art multimodal survival prediction methods would be useful for obtaining a more complete understanding of the relative performance of the proposed approach.

These limitations motivate further investigation while also defining a clear boundary for our project. Our project does not attempt to address all of these limitations. Instead, it focuses specifically on an alternative parameter-learning procedure and explicit probabilistic inference within the same latent-variable PGM.

3.10 Relationship Between the Reference Work and Our Project

The selected reference work provides the methodological foundation for our project. Our proposal explicitly states that the original multimodal PGM will first be reproduced using the available genomic and histological features, after which an EM-based parameter-learning procedure will be investigated for the same model.

The relationship between the reference work and the proposed project is summarized in Table 1.

Table 1: Comparison of the selected reference work with the proposed project.

Aspect	Selected Reference Work	Our Project
Problem	Multimodal breast cancer survival prediction using genomic and histology information.	Same multimodal breast cancer survival prediction problem.
Dataset	TCGA-BRCA with genomic and histology-derived features.	Same processed TCGA-BRCA data associated with the reference implementation.
Modalities	Genomics and histology/imaging.	Genomics and histology/imaging.
PGM Representation	Shared latent variable Z with $Z \rightarrow X$ and $Z \rightarrow Y$.	Same latent-variable PGM.
Parameter Learning	CCA-based parameter estimation, with penalized CCA variants investigated.	CCA reproduction as the baseline and EM-based parameter learning as the proposed extension.
Inference	Inference of the latent representation from multimodal observations.	Explicit evaluation of $P(Z X, Y)$, $P(Z X)$, and $P(Z Y)$, with MAP inference where applicable.
Incomplete Modalities	Not the central experimental focus.	Explicit genomic-only and histology-only experiments.
Survival Prediction	Cox proportional hazards model using the learned multimodal embeddings.	Use the learned latent representation for downstream survival prediction and compare CCA and EM approaches.
Primary Evaluation	C-index for survival prediction, together with latent-variable and additional-correlation analyses.	C-index, EM convergence behaviour, latent-state inference, and changes in survival prediction under incomplete modalities.
Project Contribution	CCA-based multimodal latent representation and survival prediction.	EM as an alternative parameter-learning method for the same PGM, with explicit latent-state and missing-modality analysis.

3.11 SOTA Position of the Proposed Work

The relationship between the selected reference work and our proposed extension can be summarized as follows:

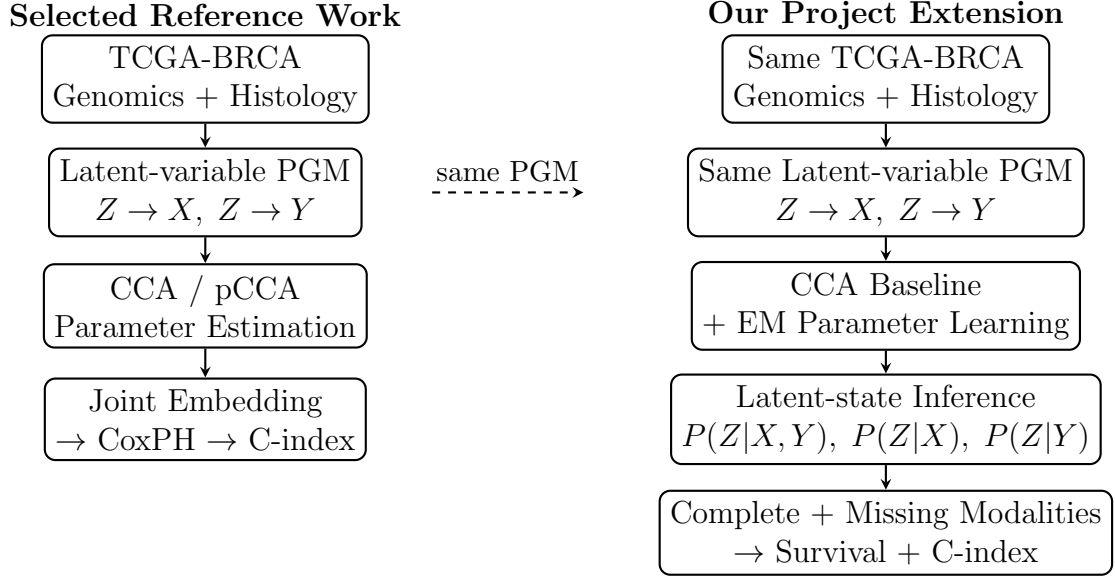


Figure 3: Initial SOTA positioning of the proposed project relative to the selected reference work. The project retains the same latent-variable PGM and uses the CCA-based approach as a baseline, while extending the learning and inference stages through EM-based parameter learning, explicit latent-state inference, and incomplete-modality experiments.

Figure 3 summarizes the initial position of our project relative to the selected reference work. The reference work provides the original latent-variable PGM and CCA-based learning pipeline. Our project retains this formulation and uses the original CCA-based approach as the baseline. The primary methodological extension is EM-based parameter learning for the same PGM.

In addition, our project explicitly treats latent-state inference as an experimental component. We will evaluate multimodal posterior inference using $P(Z | X, Y)$ and modality-specific inference using $P(Z | X)$ and $P(Z | Y)$. The project will further investigate incomplete-modality conditions by removing either genomic or histological observations and examining the resulting changes in latent-state inference and survival prediction.

Thus, the initial SOTA position of the project is not a claim that the proposed EM approach is already superior to the reference method or to all existing multimodal survival methods. Instead, the project establishes a controlled extension of the selected reference work: the same dataset and latent-variable PGM are retained, the CCA-based implementation provides the baseline, and EM-based parameter learning, explicit latent-state inference, and incomplete-modality analysis constitute the proposed extensions.

The quantitative comparison between CCA-based and EM-based learning will be established through the experiments in subsequent milestones. The SOTA position will therefore be refined as the project progresses and additional experimental evidence becomes available.

4 Baseline and Initial KPIs

Before conducting experiments, the project defines a baseline and a set of key performance indicators (KPIs) for evaluating the proposed probabilistic graphical model. Since the project focuses on the three central PGM components of representation, learning, and inference, the evaluation will not be based only on the final survival-prediction performance. Instead, the project will examine the quality of the latent representation, the behaviour of the learning procedure, and the ability of the model to infer the latent state under different observation settings.

4.1 Baseline

The existing CCA-based approach from the reference project will be used as the baseline for comparison. The baseline will first be reproduced using the processed TCGA-BRCA data and the corresponding implementation provided with the reference work.

The purpose of reproducing the baseline is to establish a reliable reference for evaluating the proposed modification. The same data configuration and evaluation settings will be used when comparing the baseline and proposed approach, wherever applicable.

The comparison will therefore consist of two approaches:

- **CCA baseline:** the existing CCA-based learning procedure used in the reference implementation to obtain a joint representation of genomic and histology-derived data.
- **Proposed approach:** an alternative parameter-learning procedure for the same latent-variable PGM, with EM investigated as the proposed extension.

The graphical structure of the PGM will remain fixed during this comparison. Therefore, the comparison will primarily focus on the effect of the parameter-learning procedure rather than introducing a different graphical structure.

The baseline comparison can therefore be represented as

CCA Baseline $\longleftrightarrow$ Proposed Parameter Learning
--

while keeping the underlying PGM structure

$$Z \rightarrow X, \quad Z \rightarrow Y$$

unchanged.

4.2 Initial Key Performance Indicators

The project defines KPIs corresponding to the three PGM components: **Representation**, **Learning**, and **Inference**. A downstream survival-prediction KPI is also included to evaluate the practical usefulness of the learned and inferred latent representation.

These KPIs are defined before experimentation so that the baseline and proposed approach can be compared using predefined evaluation criteria.

4.2.1 Representation KPIs

The representation stage concerns the latent space obtained from the genomic and histology-derived modalities. The evaluation will examine whether the learned latent representation captures information shared between the two modalities and whether the representation remains reasonably stable across the predefined data splits.

Shared and cross-modal representation. The project will examine the latent representation obtained from the two modalities to determine whether genomic and histology-derived information are represented jointly through the latent variable Z .

The representation is based on the graphical structure

$$X \leftarrow Z \rightarrow Y.$$

Therefore, the representation will be examined in terms of whether the latent space captures information that connects the two observed modalities.

Representation stability. The consistency of the learned latent representation across the predefined data splits will also be examined. A representation that changes substantially across different training splits may indicate sensitivity to the particular training data used.

This KPI will therefore provide an additional assessment of the reliability of the learned representation beyond its downstream survival-prediction performance.

4.2.2 Learning KPIs

The learning stage determines the parameters of the latent-variable PGM from the available training data. The evaluation will therefore examine whether the proposed learning procedure can reach a stable solution and whether its behaviour is practically feasible.

Convergence behaviour. For the proposed iterative parameter-learning procedure, convergence will be monitored across iterations. The change in the relevant objective or model parameters will be examined to determine whether the learning process is approaching a stable solution.

For the proposed EM-based learning procedure, the E-step and M-step will be repeated until the selected convergence criterion is satisfied.

Parameter and objective stability. The change in the model parameters or objective value between successive iterations will be monitored. This provides an indication of whether the learning procedure is stabilizing rather than continuing to make large updates.

The learning behaviour will also be examined across the predefined data splits to determine whether similar learning behaviour is observed under different training subsets.

Computational behaviour. Where practical, the computational behaviour of the baseline and proposed learning procedures will also be recorded. This may include execution time, number of iterations required for convergence, and general convergence behaviour.

The purpose of this KPI is not to perform a separate computational optimization study, but to determine whether the proposed parameter-learning procedure is practically feasible for the available BRCA data.

4.2.3 Inference KPIs

The inference stage evaluates the ability of the learned PGM to estimate the latent variable Z from the observations available for a patient.

Three observation settings will be considered:

$$p(Z \mid X, Y),$$

when both genomic and histology-derived observations are available,

$$p(Z \mid X),$$

when only genomic observations are available, and

$$p(Z \mid Y),$$

when only histology-derived observations are available.

Multimodal latent inference. The primary inference setting will use both observed modalities:

$$(X, Y) \rightarrow p(Z \mid X, Y) \rightarrow \hat{Z}_{XY}.$$

The resulting latent representation $\hat{Z}_{XY}$ will be evaluated as the representation obtained when complete multimodal information is available.

Single-modality latent inference. The project will also investigate whether the latent variable can be inferred when one modality is unavailable.

For genomic-only inference,

$$X \rightarrow p(Z \mid X) \rightarrow \hat{Z}_X.$$

For histology-only inference,

$$Y \rightarrow p(Z \mid Y) \rightarrow \hat{Z}_Y.$$

These settings provide a way to examine how the latent representation behaves when only one source of patient information is available.

Inference consistency. The inferred latent representations obtained under complete and single-modality settings will be compared to examine the effect of removing one modality.

The purpose is not to assume that the representations will be identical. Instead, the comparison will determine how much the inferred latent state changes when genomic or histology information is unavailable.

This provides an inference-oriented evaluation of the PGM rather than evaluating missing-modality settings only through downstream survival prediction.

4.2.4 Downstream Survival-Prediction KPI

Although survival prediction is not the only evaluation criterion, it provides the final application-level measure of the usefulness of the learned and inferred latent representation.

The primary downstream KPI will be the **concordance index (C-index)**.

The C-index will be used to compare the survival-prediction performance obtained from the CCA baseline and the proposed parameter-learning approach.

Where applicable, the comparison will also consider the different modality settings:

$$(X, Y), \quad X, \quad Y.$$

The C-index measures the agreement between the predicted survival risk ordering and the observed survival outcomes. A higher C-index indicates better concordance between the predicted and observed ordering of survival risk.

The C-index will therefore provide an application-level measure of whether the learned and inferred latent representation is useful for breast cancer survival prediction.

4.3 KPI Summary

The initial KPIs defined for the project are summarized in Table 2. These KPIs are intended to evaluate the PGM at multiple levels rather than reducing the evaluation to a single prediction metric.

Table 2: Initial KPIs for evaluating the proposed PGM.

Component	KPI	Purpose
Representation	Shared/cross-modal latent representation	Evaluate whether genomic and histology-derived information are captured jointly through the latent variable.
	Representation stability	Examine the consistency of the learned latent representation across predefined data splits.
Learning	Convergence behaviour	Determine whether the proposed iterative learning procedure reaches a stable solution.
	Parameter/objective stability	Monitor changes in model parameters or objective values across iterations.
	Computational behaviour	Record practical measures such as execution time and iterations required for convergence where feasible.
Inference	Multimodal inference	Evaluate latent-state inference using both genomic and histology observations, $p(Z X, Y)$.
	Genomics-only inference	Evaluate latent-state inference using genomic observations alone, $p(Z X)$.
	Histology-only inference	Evaluate latent-state inference using histology observations alone, $p(Z Y)$.
	Inference consistency	Examine how the inferred latent state changes when one modality is unavailable.
Downstream	C-index	Evaluate the usefulness of the learned/inferred latent representation for breast cancer survival prediction and compare the baseline with the proposed approach.

4.4 Success Criteria

The success of the proposed approach will not be determined by a single prediction metric. Instead, the project will consider whether the proposed method provides useful behaviour across the representation, learning, and inference stages, together with its downstream survival-prediction performance.

The proposed approach will be considered useful if it demonstrates the following:

1. a meaningful shared representation of the genomic and histology-derived modalities;
2. reasonably stable representation and learning behaviour across the predefined data splits;
3. convergence to a stable solution during parameter learning;
4. the ability to infer the latent representation using both complete and single-modality observations; and
5. competitive or improved downstream survival-prediction performance compared with the CCA baseline, as measured using the C-index.

The project therefore does not define success solely as achieving a higher survival-prediction score. The evaluation considers the complete PGM pipeline:

Representation $\rightarrow$ Learning $\rightarrow$ Inference $\rightarrow$ Survival Prediction

This evaluation framework will be established before experimentation so that the baseline and proposed approach can be compared using consistent data splits, inference settings, and evaluation criteria.

5 Experimental Plan, Team Responsibilities and Timeline

The project will be carried out through a milestone-based experimental plan. The work will progress from establishing the reference baseline to implementing the proposed extension, evaluating latent-state inference, and finally comparing the approaches using the predefined KPIs.

5.1 Experimental Plan

The overall experimental workflow is organized as follows:

Baseline $\rightarrow$ Proposed Learning $\rightarrow$ Inference $\rightarrow$ Evaluation

The existing CCA-based approach will first be reproduced and validated using the provided TCGA-BRCA data. The proposed parameter-learning extension will then be implemented using the same latent-variable PGM. The learned model will subsequently be evaluated for multimodal and single-modality latent-state inference, followed by incomplete-modality experiments. Finally, the baseline and proposed approaches will be compared using the Representation, Learning, Inference, and downstream C-index KPIs defined in Section 4.

5.2 Team Responsibilities

The project responsibilities are distributed across all four team members, with each member contributing to both implementation and evaluation. Each member will take primary responsibility for one area while participating in the remaining stages of the project.

Table 3: Team responsibilities.

Team Member	Primary Responsibilities
Dhanya Achari	PGM formulation and representation; dataset preparation and verification; participation in baseline reproduction and result analysis.
Shifataj Shaikh	Reference-method/CCA baseline implementation; parameter-learning experiments; participation in inference and result analysis.
Parth	Proposed parameter-learning implementation; latent-state inference for multimodal and single-modality settings; participation in baseline evaluation.
Visruth	Incomplete-modality experiments; KPI-based evaluation and comparison; participation in learning and inference experiments.

All team members will contribute equally to literature review, experiment validation, result interpretation, report writing, and final presentation.

5.3 Timeline

The project will be completed through four milestones. Each milestone has a defined objective and expected deliverable.

Table 4: Project timeline and milestone-wise deliverables.

Milestone	Planned Work	Expected Deliverable
M1	Problem definition, reference study, PGM formulation, baseline selection, KPI definition, and project planning.	Finalized project proposal and experimental plan.
M2	Dataset verification and preprocessing; reproduction and validation of the CCA-based baseline.	Verified dataset pipeline and baseline results.
M3	Implementation of the proposed parameter-learning extension; multi-modal and single-modality latent-state inference.	Proposed implementation and intermediate results.
M4	Incomplete-modality experiments; KPI-based comparison of the baseline and proposed approach; final analysis.	Final experimental results, report, and presentation.

The milestone structure ensures that the project progresses incrementally from baseline reproduction to the proposed extension and its evaluation. Each milestone will provide the foundation for the subsequent stage while allowing the results to be validated before proceeding further.

References

- [1] V. Subramanian, T. Syeda-Mahmood, and M. N. Do, “Modelling-based joint embedding of histology and genomics using canonical correlation analysis for breast cancer survival prediction,” *Artificial Intelligence in Medicine*, vol. 149, Art. no. 102787, 2024. doi: 10.1016/j.artmed.2024.102787.

Research Paper – IBM Research: IBM Research Publication Page

Authors’ Code and Processed Dataset: GitHub Repository

Our Group GitHub Repository: CSE-516–PGM Healthcare BRCA